Aden Hilderbrand

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Education

Embry-Riddle Aeronautical University — Anticipated May 2027

B.S. Cyber Intelligence & Security — Prescott, AZ

GPA 3.86/4.0 • Dean's List: Fall 2024, Spring 2025, Fall 2025 • Red Hat Certified System Administrator (RHCSA)

Experience

Embedded Cyber Security Software Engineer — Honeywell Aerospace — May 2026 – Aug. 2026

Phoenix, AZ

- Hardened embedded software security posture on aerospace flight systems, designing and implementing API endpoints on a lighttpd/FastCGI stack for inter-process and inter-device communication.
- Built iOS companion tooling in Swift bridging proprietary avionics systems and mobile devices, connecting the FastCGI API layer to a BLE GATT server and HTTPS file-transfer channel for secure on-device data exchange.
- Developed embedded Bluetooth functionality and integrated hardware-received ADS-B and GPS data feeds into proprietary flight-adjacent software; contributed directly to production codebases in C++ and Bash.
- Tracked requirements and deliverables through JIRA and DOORS and maintained documentation in Confluence; ran regular sync meetings to keep sprint objectives on track.

C++, Bash, Swift, lighttpd/FastCGI, BLE/GATT, ADS-B, GPS, JIRA, DOORS, Confluence

Ground School Data Manager — Embry-Riddle Aeronautical University — Oct. 2025 – May 2026

Prescott, AZ

- Maintained FERPA-compliant records for 300+ students per semester, enforcing data integrity, system accessibility, and compliance across proprietary academic platforms.
- Administered SharePoint to streamline document workflows and enforce authorized-only access to confidential exam and student record data; supported faculty on updated access-control procedures.

SharePoint Administration, FERPA Compliance, Access Control

Front of Store Specialist & Trainer — Target Corp. — Oct. 2021 – Jan. 2026

Phoenix, AZ

- Led onboarding and training for a dozen+ new hires, developing instructional procedures that accelerated ramp-up and improved team consistency; served as escalation point for roughly 400 customer interactions per week.

Projects

Honeywell / ERAU Undergraduate Research Project — Oct. 2025 – May 2026

- Engineered encrypted wireless data transfer pipelines and key management schemes over BLE/WiFi for a high-constraint embedded Linux platform (NXP RDB2) targeting advanced air mobility systems.
- Applied enterprise-grade cryptographic standards to resource-constrained hardware in a regulated aviation domain, coordinating directly with Honeywell engineers and ERAU faculty.

Embedded Linux, NXP RDB2, BLE/WiFi, Key Management, Applied Cryptography

CyberAUTO Summit — Automotive Penetration Testing

- Conducted hands-on penetration testing against real-world automotive systems on a 10-person team, tapping CAN bus traffic (SocketCAN, SavvyCAN, python-can, OBD-II) and assessing EV/EVSE charging infrastructure.

- Exploited onboard and infrastructure web services (Burp Suite, pwntools, Scapy, XSS/SQLi) and reverse-engineered hardware down to the chip (JTAG, SOIC-8, chip-off extraction).
- Pulled forensic artifacts with Berla iVe and probed RF/BLE attack surface via SDR tooling (GNURadio, HackRF, nRF Connect).

SocketCAN, python-can, OBD-II, EV/EVSE, Burp Suite, JTAG, Berla iVe, GNURadio, HackRF, GDB, Ghidra

TracerFire13 — Cyber Threat Analyst 6th of 34 teams

- Identified and attributed adversary activity in a live enterprise cyber-defense exercise under a compressed operational timeline.
- Leveraged endpoint detection, cloud infrastructure analysis, and network packet capture tooling to trace realistic attacker behavior.

Autopsy, Velociraptor, Microsoft Azure, Malcolm

Fail2ban Defensive Learning Lab — Published Lab Module

- Authored a publication-ready lab on SSH brute-force protection, accepted into *Mastering Enterprise Networks, 3rd Edition* (ERAU / Eagle Publications).
- Built a three-machine GNS3 + VirtualBox environment and wrote instructional content covering Fail2ban, iptables automated banning, and live Hydra brute-force simulation.

Fail2ban, iptables, Hydra, GNS3, VirtualBox, SSH Hardening

NSA Codebreaker Challenge 2024

- Advanced through a multi-stage national cryptanalysis and reverse-engineering competition requiring progressive exploitation of realistic, scenario-driven problems.
- Applied binary analysis, applied cryptography, and scripted automation to advance through increasingly complex challenge tiers.

GDB, Ghidra, Python, Applied Cryptography

Encrypted Chat Application

- Built a secure, socket-based chat application in Python implementing RSA public-key encryption for end-to-end confidentiality.
- Used multithreading to handle concurrent client connections without blocking I/O.

Python, RSA, Socket Programming, Multithreading

Signal Metadata Exploit — RTT Side-Channel Research

- Implemented an exploit of the packet round-trip-time metadata leak in Signal Messenger documented in the *Careless Whisper* paper (arXiv:2411.11194).
- Inferred a target's device activity and session state from delivery-receipt timing alone, with no notification generated on the victim's device.

signal-cli, socat, Bash, Side-Channel Analysis

CyberEye Club — Vice President, Embry-Riddle Aeronautical University

- Coordinated technical workshops, competition team preparation, and skill-development initiatives for ERAU's cybersecurity student organization.
- Bridged classroom theory with applied offensive and defensive security practice for a competition-ready membership.

Leadership, Event Coordination, Technical Mentorship

Additional Skills

Languages: Python, C, C++, Go, Bash/Shell, PowerShell, JavaScript, Swift
Systems: Linux (Ubuntu, Kali, RHEL), Windows, UNIX Administration, Docker, Embedded Linux
Security Tools: Wireshark, Ghidra, GDB, Autopsy, FTK Imager, Volatility3, Velociraptor, Malcolm, Hydra, Fail2ban
Networking & Crypto: iptables, SSH Hardening, RF/Wireless Analysis, BLE/WiFi, Asymmetric/Symmetric Cryptography, Key Management, Secure Boot